



CONFIDENTIAL

SPARK · GROWTH ANALYTICS

data analyst intern.

take-home assignment. about four hours of work, forty-eight hours to send it back. rough is fine — we care about how you think.

ROLE

data analyst intern

PRODUCT

Rasoi · practice product

TIME

~4 hours · 48 hours to submit

SPARK · TAKE-HOME

before you start

Propel builds and runs a portfolio of consumer apps for indian users, almost all of it grown with paid media. analytics here decides where the money goes next month — so this work ends in a recommendation, not a dashboard.

this assignment is about **four hours of work** and you have **48 hours** from when you receive it. rough is fine — we are reading your judgement, not your formatting.

the product you're working with

Rasoi is a practice product. we invented it for this assignment — it is not one of ours and there is nothing to look up. treat everything below as true.

an android cooking app for indian users: regional-language recipes, step-by-step video, shopping lists. a ₹1 trial purchase starts it; 24 hours later the real plan activates on its own — ₹199 a month or ₹1,199 a year, decided by a backend experiment at the moment of trial. acquisition is entirely paid: meta, google and sharechat, six live campaigns between them. all money below is ex-gst.

ONE THING WORTH KNOWING

if something about the data or the brief isn't clear, ask us. asking a good question is a point in your favour, not a penalty. guessing silently is the thing we're watching for.

AND ONE MORE

every role here runs on ai agents — Claude, OpenAI models, whatever is working that month. using them on this assignment is expected, not tolerated. task 6 asks you to show your working.

the file

rasoi_growth_data.csv is attached to the email you got this on. 540 rows — one row per campaign per day, 1 june to 29 august 2026.

COLUMN	WHAT IT IS
date	day of activity, ist
channel	meta, google or sharechat
campaign	six values, e.g. mt_lookalike_3pct
spend_inr	media spend, ex-gst
impressions, clicks, installs	the usual, attributed
trials	₹1 trial purchases started that day
renewals	real plans charged that day, 24h after the trial
annual_renewals	the slice of renewals on the ₹1,199 plan
revenue_inr	gross revenue booked that day, ₹1 trials included
refunds_inr	refunds and chargebacks booked that day

task 1 — trust the file · 45 MIN

before you analyse anything, satisfy yourself the data means what it says.

- profile it: completeness, ranges, anything impossible or implausible.
- **there is one real problem in this file.** find it, say what you think happened, say which rows it touches, and say exactly how you handled it — dropped, flagged, interpolated, left alone. how you handled it matters more than what you called it.
- anything else you'd want from an engineer before putting real money behind your answer.

task 2 — the six campaigns · 60 MIN

one row per campaign, full period. at minimum:

```
cost per install:
cost per trial:
cac per paying user:
trial to paid %:
annual plan mix:
first-month roas, net of refunds:
```

THE ONE TRAP WE'LL WARN YOU ABOUT

a renewal on any given day belongs to the trial started the day before. averaging same-day ratios across a period where spend is moving gives you a number that is wrong in a direction you should be able to name.

task 3 — which ones actually work · 30 MIN

- rank the six by cost per trial. rank them again by the metric you actually believe. does the order change?
- if it does — in plain language, is the cheap-looking campaign a trap or not? a founder should follow this without seeing your query.
- nominate **one** metric the team should steer on week to week, and defend it in three sentences. we disagree about this internally, so argue it rather than guessing what we want to hear.

task 4 — split the money · 45 MIN

next month's paid budget is **₹1 crore**. today it's split roughly in proportion to the last ninety days and nobody has checked whether that's right. give us a table: proposed spend, expected trials, expected paying users, expected first-month revenue, per campaign.

- list your assumptions. diminishing returns at higher spend is the one people forget — decide whether you believe in it and say so either way.
- name the one campaign you would **kill**, and the evidence that would prove you wrong.
- one headline number for the whole plan. say which number, and why that one.

task 5 — one chart · 20 MIN

make **one** chart — not a dashboard — that a founder could look at for ten seconds and know what to do. any tool.

underneath it, write one sentence saying what it is telling them to do. if you can't write that sentence, it's the wrong chart.

task 6 — how you used ai · 15 MIN

- which tools you used, and for what — queries, cleaning, charting, writing.
- your single best prompt, pasted exactly as you sent it.
- **one thing the model got wrong** — a query that silently returned the wrong grain, a confident claim the data doesn't support, a chart that misled — how you caught it and what you did. an answer saying it was right about everything reads as either no ai or no checking.

how to send it back

three things, replying to the same email thread you got this on:

- your working — notebook, .py, .sql or a sheet. it should recompute from the csv.
- one page as a doc or pdf: the recommendation first, the evidence second, the caveats last.
- the chart, as an image or inside that page.

no machine learning. there is nothing in this file that needs a model, and reaching for one counts against you.

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